

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Exam #3

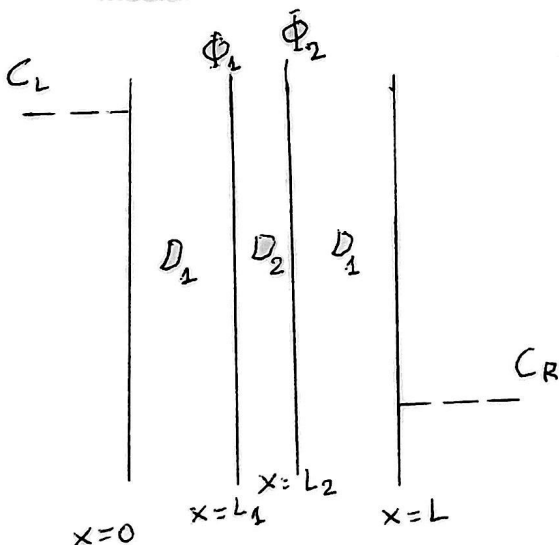
Duration: 70 minutes --- November 25, 2025

All Problems must be done

Problem 1: (30 points)

Consider steady state diffusion of the species A through three media arranged as shown in the attached diagram. Assume that diffusion is one-dimensional. At $x = 0$, $C_{A1} = C_L$, and at $x = L$, $C_{A3} = C_R$. In medium 2, there is a zeroth order chemical reaction through which the species A is consumed. Develop an expression for the concentration profiles across the three media.

(Bonus 10 points) Develop an expression for the steady state flux across the three media.



Balance Equations:

$$D_1 \frac{d^2 C_{A1}}{dx^2} = K_1$$

$$D_2 \frac{d^2 C_{A2}}{dx^2} = 0$$

$$D_1 \frac{d^2 C_{A3}}{dx^2} = K_1$$

$$\frac{d^2 C_{A1}}{dx^2} = \frac{K_1}{D_1}$$

$$\frac{dC_{A1}}{dx} = \frac{K_1}{D_1} x + A$$

$$C_{A1} = \frac{K_1}{2D_1} x^2 + Ax + B$$

$$\frac{d^2 C_{A2}}{dx^2} = 0 \rightarrow \frac{dC_{A2}}{dx} = D, \quad C_{A2} = Dx + E$$

$$\frac{d^2 C_{A3}}{dx^2} = \frac{K_1}{D_1} \rightarrow \frac{dC_{A3}}{dx} = \frac{K_1}{D_1} x + G, \quad C_{A3} = \frac{K_1}{2D_1} x^2 + Gx + H$$

Using B.C.1: $C_L = B$

Using B.C.2: $C_R = \frac{K_1}{2D_1} L^2 + GL + H \rightarrow H = C_R - \frac{K_1}{2D_1} L^2 - GL$

Using B.C.3: $\frac{K_1}{D_1} L_1 + A = D$

Using B.C.4: $D = \frac{K_1}{D_1} L_2 + G \rightarrow A - G = \frac{K_1}{D_1} (L_2 - L_1)$

Boundary Conditions:

B.C.1: $C_{A1}(x=0) = C_L$

B.C.2: $C_{A3}(x=L) = C_R$

B.C.3: $\left. \frac{dC_{A1}}{dx} \right|_{x=L_1} = \left. \frac{dC_{A2}}{dx} \right|_{x=L_1}$

B.C.4: $\left. \frac{dC_{A2}}{dx} \right|_{x=L_2} = \left. \frac{dC_{A3}}{dx} \right|_{x=L_2}$

B.C.5: $\frac{C_{A2}(x=L_1^+)}{C_{A1}(x=L_1^-)} = \phi_1$

B.C.6: $\frac{C_{A3}(x=L_2^+)}{C_{A2}(x=L_2^-)} = \phi_2$

Using B.C.5:
$$\frac{DL_1 + E}{\frac{K_1}{2D_1}L_1^2 + AL_1 + C_L} = \phi_1$$

Using B.C.6:
$$\frac{\frac{K_1}{2D_1}L_2^2 + GL_2 + H}{DL_2 + E} = \phi_2$$

Steady State flux:

$$\frac{dC_{A1}}{dx} + \frac{dC_{A2}}{dx} + \frac{dC_{A3}}{dx} = \frac{K_1}{D_1}x + A + D + \frac{K_1}{D_1}x + G$$

$$\frac{dC_A}{dx} = \frac{2K_1}{D_1}x + A + G + D$$

$$D = \frac{K_1}{D_1}L_2 + G$$

$$A = \frac{K_1}{D_1}(L_2 - L_1) + G$$

$$= \frac{2K_1}{D_1}x + \frac{K_1}{D_1}(L_2 - L_1) + G + \frac{K_1}{D_1}L_2 + G + G$$

$$\frac{dC_A}{dx} = \frac{K_1}{D_1}(x + L_2 - L_1 + L_2) + 3G$$

$$\boxed{\frac{dC_A}{dx} = \frac{K_1}{D_1}(x + 2L_2 - L_1) + 3G}$$

Problem 2: (70 points)

Consider the reversible conversion of species A into B through the stoichiometric reactions $A \xrightleftharpoons[k_{-1}]{k_1} B$ in a stationary liquid film of thickness L . Determine C_A and C_B throughout the film, if

a) (35 Points) the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentrations of A and B at $y = 0$ are specified as C_{A0} and C_{B0} . The forward and reverse homogeneous reactions follow zeroth order kinetics

b) (35 Points) the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentrations of A and B at $y = 0$ are specified as C_{A0} and C_{B0} . The forward and reverse homogeneous reactions follow zeroth and first order kinetics, respectively

a) Balance Equation.

$$D_{AB} \frac{d^2 C_A}{dy^2} = k_1$$

$$D_{BA} \frac{d^2 C_B}{dy^2} = k_{-1}$$

Boundary Conditions

$$B.C.1: C_A(y=0) = C_{A0}$$

$$B.C.2: C_A(y=L) = 0$$

$$B.C.3: C_B(y=0) = C_{B0}$$

$$B.C.4: C_B(y=L) = 0$$

$$\frac{d^2 C_A}{dy^2} = \frac{k_1}{D_{AB}} \rightarrow \frac{dC_A}{dy} = \frac{k_1}{D_{AB}} y + D$$

$$C_A = \frac{k_1}{2D_{AB}} y^2 + Dy + E$$

$$C_A = \frac{k_1}{2D_{AB}} y^2 - \frac{k_1}{2D_{AB}} Ly - \frac{C_{A0}}{L} y + C_{A0}$$

$$\frac{d^2 C_B}{dy^2} = \frac{k_{-1}}{D_{BA}} \rightarrow \frac{dC_B}{dy} = \frac{k_{-1}}{D_{BA}} y + F$$

$$C_B = \frac{k_{-1}}{2D_{BA}} y^2 + Fy + G$$

$$C_B = \frac{k_{-1}}{2D_{BA}} y^2 - \frac{k_{-1}}{2D_{BA}} Ly - \frac{C_{B0}}{L} y + C_{B0}$$

Using B.C.1:

$$E = C_{A0}$$

Using B.C.2:

$$0 = \frac{k_1}{2D_{AB}} L^2 + DL + C_{A0}$$

$$D = -\frac{k_1}{2D_{AB}} L - \frac{C_{A0}}{L}$$

Using B.C.3:

$$G = C_{B0}$$

Using B.C.4:

$$0 = \frac{k_{-1}}{2D_{BA}} L^2 + FL + C_{B0}$$

$$F = -\frac{k_{-1}}{2D_{BA}} L - \frac{C_{B0}}{L}$$

b) Balance Equations

$$D_{AB} \frac{d^2 C_A}{dy^2} = k_1$$

$$D_{BA} \frac{d^2 C_B}{dy^2} = k_1 C_B$$

Boundary Conditions

$$\text{B.C.1: } C_A(y=0) = C_{A0}$$

$$\text{B.C.2: } C_B(y=0) = C_{B0}$$

$$\text{B.C.3: } C_A(y=L) = 0$$

$$\text{B.C.4: } C_B(y=L) = 0$$

C_A will follow same integration as above to yield:

$$C_A = \frac{k_1}{2D_{AB}} y^2 - \frac{k_1}{2D_{AB}} Ly - \frac{C_{A0}}{L} y + C_{A0}$$

$$\frac{d^2 C_B}{dy^2} = \frac{k_1}{D_{BA}} C_B \rightarrow \frac{d^2 C_B}{dy^2} - \frac{k_1}{D_{BA}} C_B = 0$$

$$r^2 - \frac{k_1}{D_{BA}} r = 0$$

$$r = -\sqrt{\frac{k_1}{D_{BA}}}$$

$$C_B = G e^{\sqrt{\frac{k_1}{D_{BA}}} y} + F e^{-\sqrt{\frac{k_1}{D_{BA}}} y}$$

$$C_B = G \sinh\left(\sqrt{\frac{k_1}{D_{AB}}} y\right) + F \cosh\left(-\sqrt{\frac{k_1}{D_{AB}}} y\right)$$

Using B.C.2:

$$C_{B0} = G \sinh(0) + F \cosh(0)$$

$$C_{B0} = F$$

Using B.C.4: $0 = G \sinh\left(\sqrt{\frac{k_1}{D_{AB}}} L\right) + C_{B0} \cosh\left(-\sqrt{\frac{k_1}{D_{AB}}} L\right)$ let's call $\sqrt{\frac{k_1}{D_{BA}}} = \lambda$

$$G = -\frac{C_{B0} \cosh(\lambda L)}{\sinh(\lambda L)}$$

$$C_B = -\frac{C_{B0} \cosh(-\lambda L)}{\sinh(\lambda L)} \sinh(\lambda y) + C_{B0} \cosh(-\lambda y)$$